PROJECT REPORT

Title

Implementation of Access Control Lists (ACL) in Ubuntu Linux

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1. Aim

To study and implement Access Control Lists (ACLs) in Ubuntu Linux for providing fine-grained access control to files and directories, and to understand how ACLs extend traditional Linux file permissions.

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2. Objective

1. To understand the concept of Access Control Lists (ACLs).
2. To install and configure ACL utilities in Ubuntu.
3. To assign different permissions to multiple users on the same file or directory.
4. To implement default ACLs for automatic permission inheritance.
5. To verify and manage ACL settings using Linux commands.

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3. Introduction

Linux traditionally uses three permission categories:

- Owner (User)
- Group
- Others

These permissions are represented using read (r), write (w), and execute (x) rights. However, in many real-world situations, administrators need more flexible permission management.

Access Control Lists (ACLs) provide an extended permission mechanism that allows assigning permissions to specific users and groups without changing file ownership or group membership.

ACLs are widely used in enterprise environments where multiple users require different levels of access to shared resources.

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4. Theory

What is ACL?

An Access Control List (ACL) is a list of permissions attached to an object such as a file or directory. ACLs allow administrators to define permissions for multiple users and groups beyond the traditional Linux permission model.

Types of ACL

1. Access ACL

Controls permissions for existing files and directories.

Example:

User Alice can have read-write permissions while User Bob has only read permissions.

2. Default ACL

Applied to directories.

Any newly created file or subdirectory inside the directory automatically inherits the specified ACL permissions.

Advantages of ACL

- Fine-grained access control
- Easier management of shared resources
- Reduced need for multiple groups
- Increased security
- Better collaboration among users

ACL Commands

setfacl

Used to set ACL permissions.

Syntax:

setfacl [options] file_name

getfacl

Used to display ACL permissions.

Syntax:

getfacl file_name

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5. System Requirements

Hardware Requirements

- Processor: Intel Core i3 or higher
- RAM: 4 GB minimum
- Hard Disk: 20 GB free space

Software Requirements

- Ubuntu Linux (20.04/22.04 or later)
- ACL Package
- Terminal

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6. Procedure

Step 1: Update Package Repository

```
sudo apt update
```

Step 2: Install ACL Package

```
sudo apt install acl
```

Verify installation:

```
which setfacl
which getfacl
```

Step 3: Create Users

Create two users:

```
sudo adduser alice
sudo adduser bob
```

Step 4: Create Shared Directory

```
sudo mkdir /shared
```

Assign basic permissions:

```
sudo chmod 770 /shared
```

Step 5: Set ACL Permissions

Provide full permissions to Alice:

```
sudo setfacl -m u:alice:rwx /shared
```

Provide read and execute permissions to Bob:

```
sudo setfacl -m u:bob:rx /shared
```

Step 6: View ACL Settings

```
getfacl /shared
```

Expected Output:

```
user::rwx
user:alice:rwx
user:bob:r-x
group::rwx
mask::rwx
other::---
```

Step 7: Configure Default ACLs

Set inheritance permissions:

```
sudo setfacl -d -m u:alice:rwx /shared
sudo setfacl -d -m u:bob:rx /shared
```

Verify:

```
getfacl /shared
```

Step 8: Test Permissions

Login as Alice

```
su - alice
touch /shared/file1
echo "Hello World" > /shared/file1
```

Login as Bob

```
su - bob
cat /shared/file1
```

Expected Result:

- Bob can read the file.
- Bob cannot modify or delete the file.

Attempt:

```
echo "Test" >> /shared/file1
```

Permission should be denied.

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7. Screenshots

Insert screenshots of the following:

Screenshot 1

Updating package repository

```
sudo apt update
```

Screenshot 2

Installing ACL package

```
sudo apt install acl
```

Screenshot 3

Creating users Alice and Bob

```
sudo adduser alice
sudo adduser bob
```

Screenshot 4

Creating shared directory

```
sudo mkdir /shared
```

Screenshot 5

Setting ACL permissions

```
setfacl -m u:alice:rwx /shared
setfacl -m u:bob:rx /shared
```

Screenshot 6

Displaying ACL information

```
getfacl /shared
```

Screenshot 7

Creating file as Alice

```
touch /shared/file1
```

Screenshot 8

Reading file as Bob

```
cat /shared/file1
```

Screenshot 9

Permission denied while Bob attempts modification

```
echo "Test" >> /shared/file1
```

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8. Observations

User	Read	Write	Execute
Alice	Yes	Yes	Yes
Bob	Yes	No	Yes

The ACL permissions were successfully enforced according to the configuration.

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9. Result

The implementation of Access Control Lists (ACLs) in Ubuntu Linux was successfully completed. Different permission levels were assigned to users Alice and Bob on the shared directory. ACLs provided flexible and secure access control beyond traditional Linux permissions.

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10. Conclusion

The project demonstrated the practical implementation of Access Control Lists (ACLs) in Ubuntu Linux. ACLs enable administrators to assign customized permissions to individual users and groups without altering ownership structures. The experiment verified that ACLs provide enhanced flexibility, security, and resource-sharing capabilities in multi-user environments. Therefore, ACLs are an effective solution for managing file and directory permissions in modern Linux systems.

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11. References

1. Ubuntu Documentation
2. Linux Manual Pages (setfacl, getfacl)
3. Linux System Administration Guide
4. Ubuntu ACL Package Documentation
5. GNU/Linux Security Administration Resources